E-Commerce Customer Behaviour & Sales Analytics

Tech Stack: Hadoop | Hive | MapReduce | PySpark | Cassandra | MongoDB

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The Retail Analytics Bottleneck

Traditional single-machine tools fail as transaction logs scale. Retailers struggle to answer:



Which products, times, and regions drive the most revenue?



Who are the high-value, loyal, or at-risk customers?



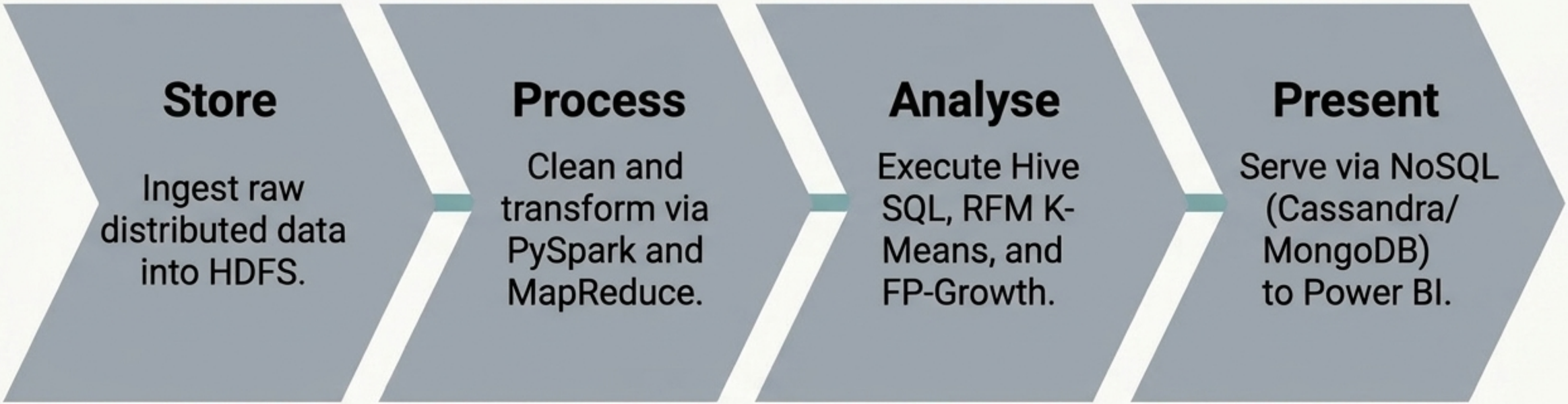
Which products are frequently purchased together?



How impactful are product cancellations and returns?

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Project Objectives: The Data Pipeline



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Alignment with Global Goals



SDG 8
Decent Work and Economic Growth

- Improves retail inventory planning and resource allocation.
- Optimizes marketing spend through targeted customer insights.
- Drives productivity and sustainable growth for mid-sized retailers.

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Raw Material: UCI Online Retail Dataset

Period: Dec 2010 – Dec 2011 | Size: ~541,909 rows

Attribute	Description
InvoiceNo	6-digit invoice number
StockCode	Product (item) code
Description	Product name
Quantity	Units per transaction
InvoiceDate	Date and time
UnitPrice	Price per unit
CustomerID	5-digit identifier
Country	Customer's country



Data Quality Hurdles

- ~25% missing CustomerIDs.
- Cancelled invoices present (negative quantities).
- Zero or negative unit prices recorded.
- Duplicate transaction rows identified.

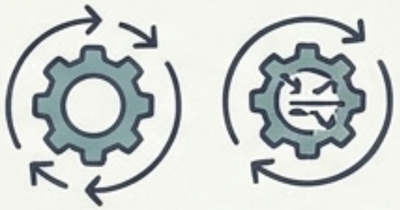
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Distributed Technology Stack



Hadoop HDFS

Distributed raw and cleaned data storage.



PySpark / MapReduce

Batch cleaning and feature engineering.



Apache Hive

SQL-style analytical warehouse over HDFS.



Cassandra

Wide-column NoSQL for time-series queries.



MongoDB

Document NoSQL for flexible customer profiles.

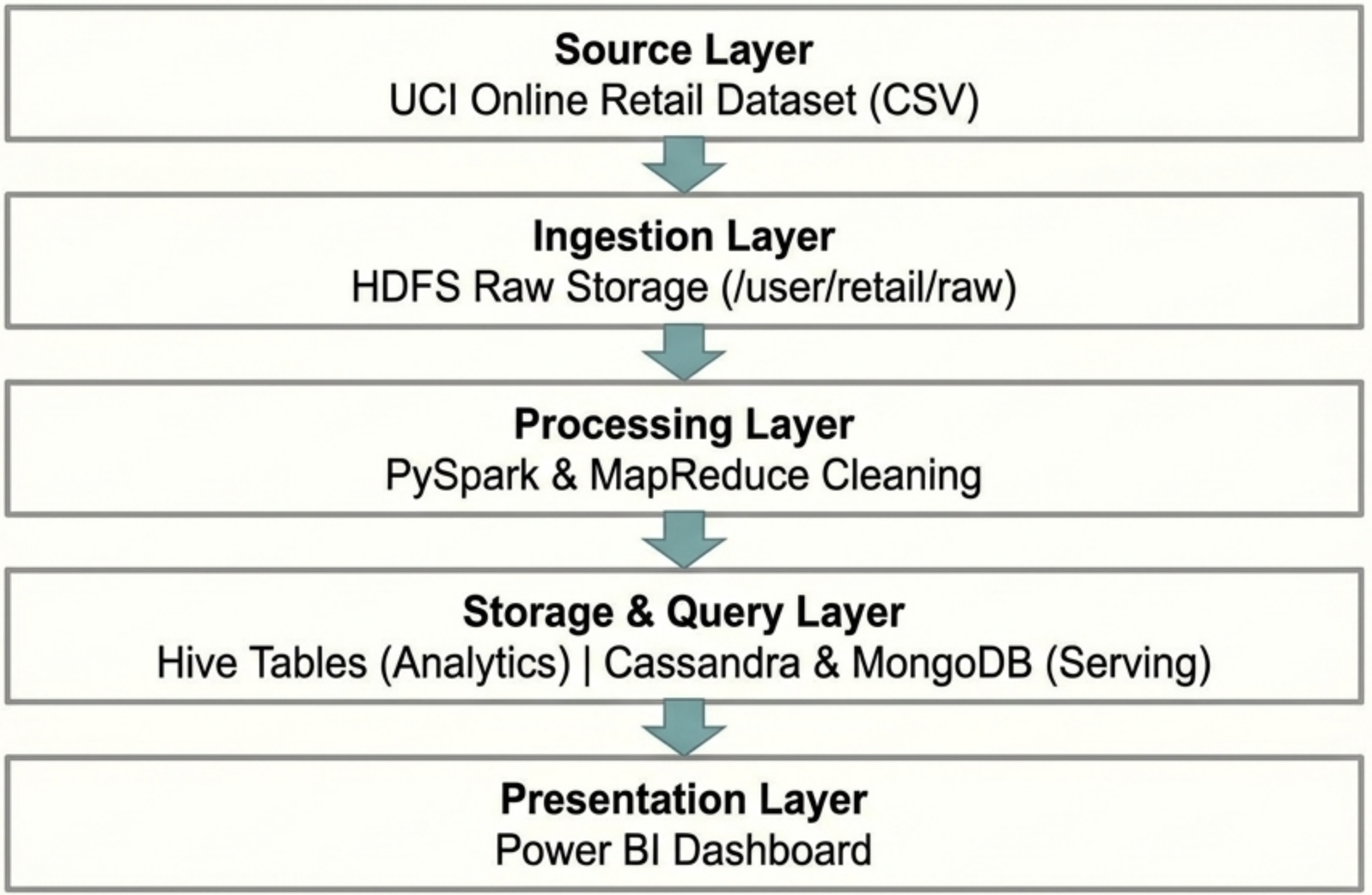


Power BI

Interactive analytical dashboard presentation.

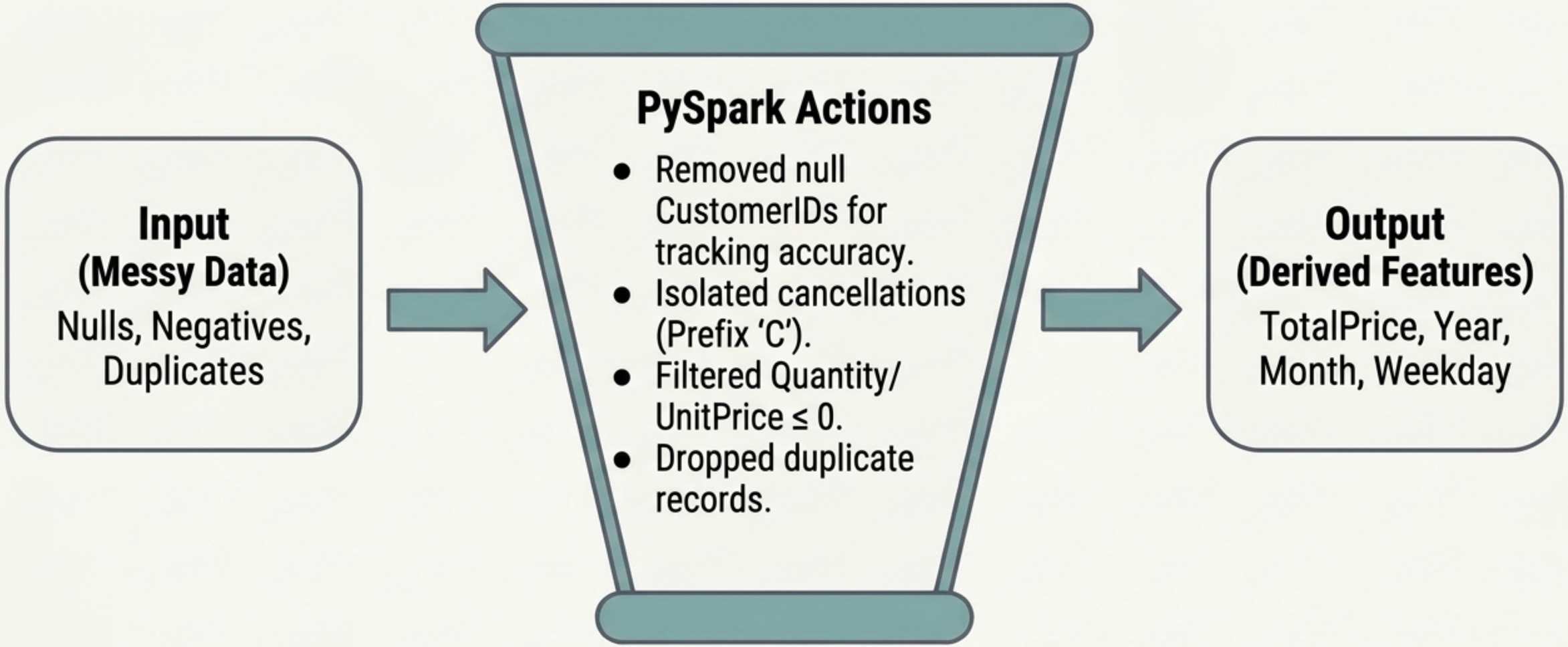
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5-Layer System Architecture



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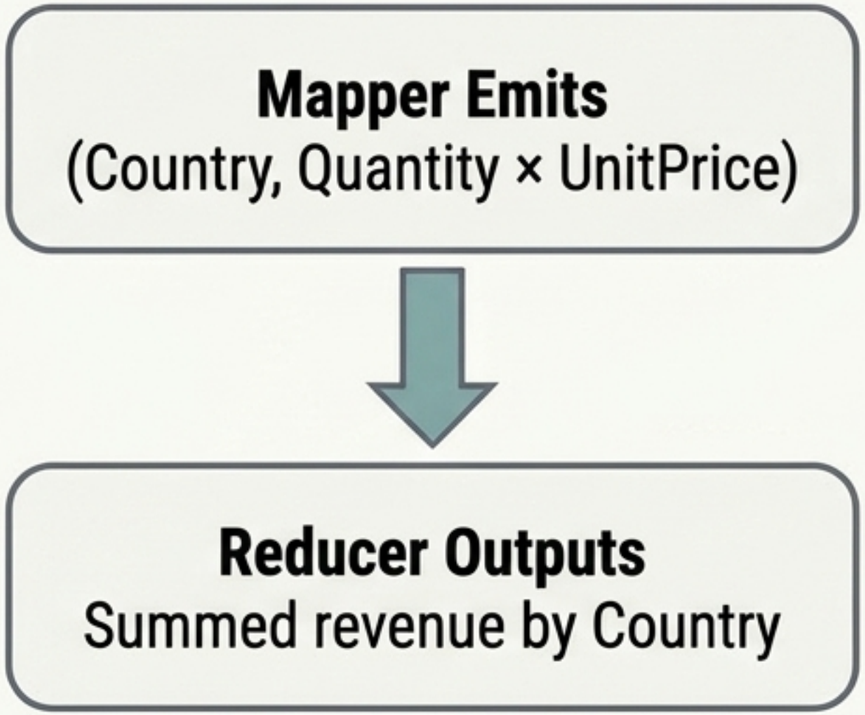
Data Transformation Pipeline



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Distributed Aggregation: MapReduce & Hive

MapReduce Flow



Hive Analytics (Snippet)

```
CREATE EXTERNAL TABLE retail_clean...

SELECT year, month, SUM(TotalPrice)
FROM retail_clean
GROUP BY year, month;
```

Enables scalable, SQL-style aggregations over massive data.

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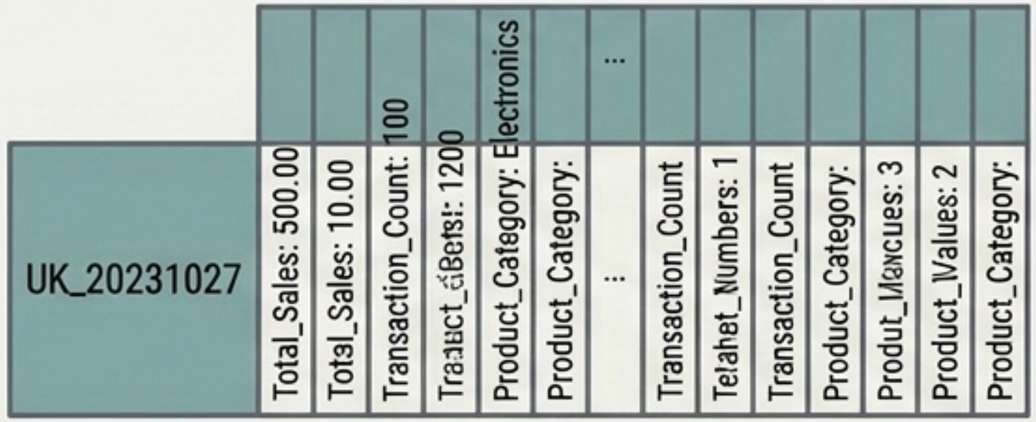
Dual NoSQL Serving Layer

Cassandra

Paradigm: Wide-Column Store

Design Focus: Query-driven for fast read access.

Stored Data: Time-series sales by country and date.

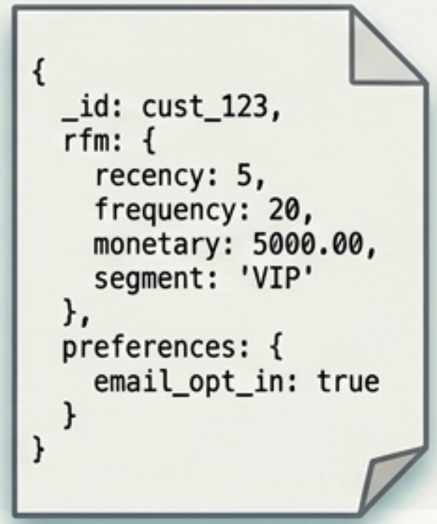


MongoDB

Paradigm: Document Store

Design Focus: Schema-less design for flexible profiles.

Stored Data: Segmented customer RFM profiles.



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RFM Customer Segmentation (K-Means)

Methodology: RFM scaled via StandardScaler; optimal k chosen via elbow method

Champions
Recent, frequent, high spend ➔ Loyalty rewards, early access

Loyal
Regular, moderate spend ➔ Upsell and cross-sell

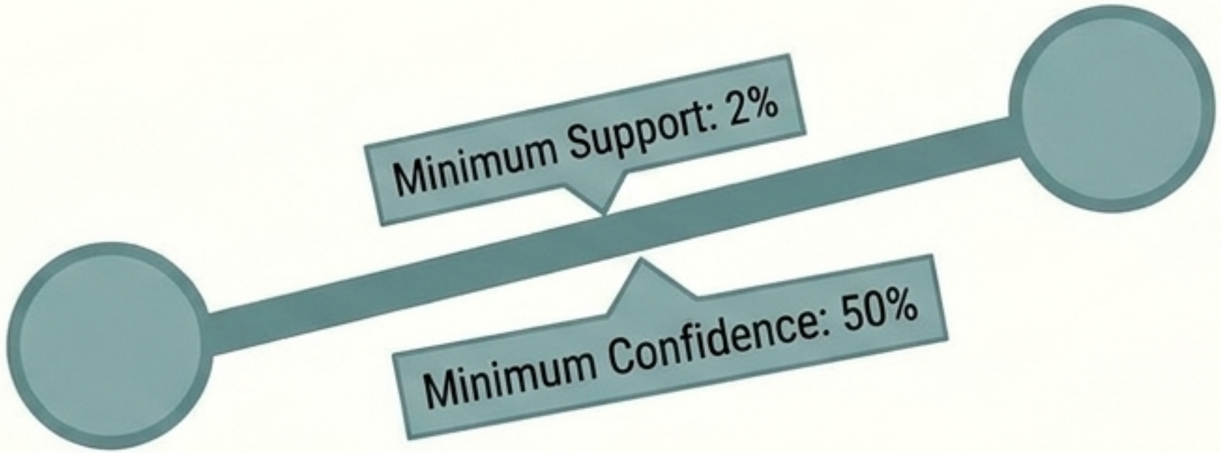
At-Risk
Previously active, now inactive ➔ Win-back offers

Lost
Low recency and frequency ➔ Low-cost reactivation

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Market Basket Analysis (FP-Growth)

Algorithm: PySpark MLlib FP-Growth on invoice-level baskets.

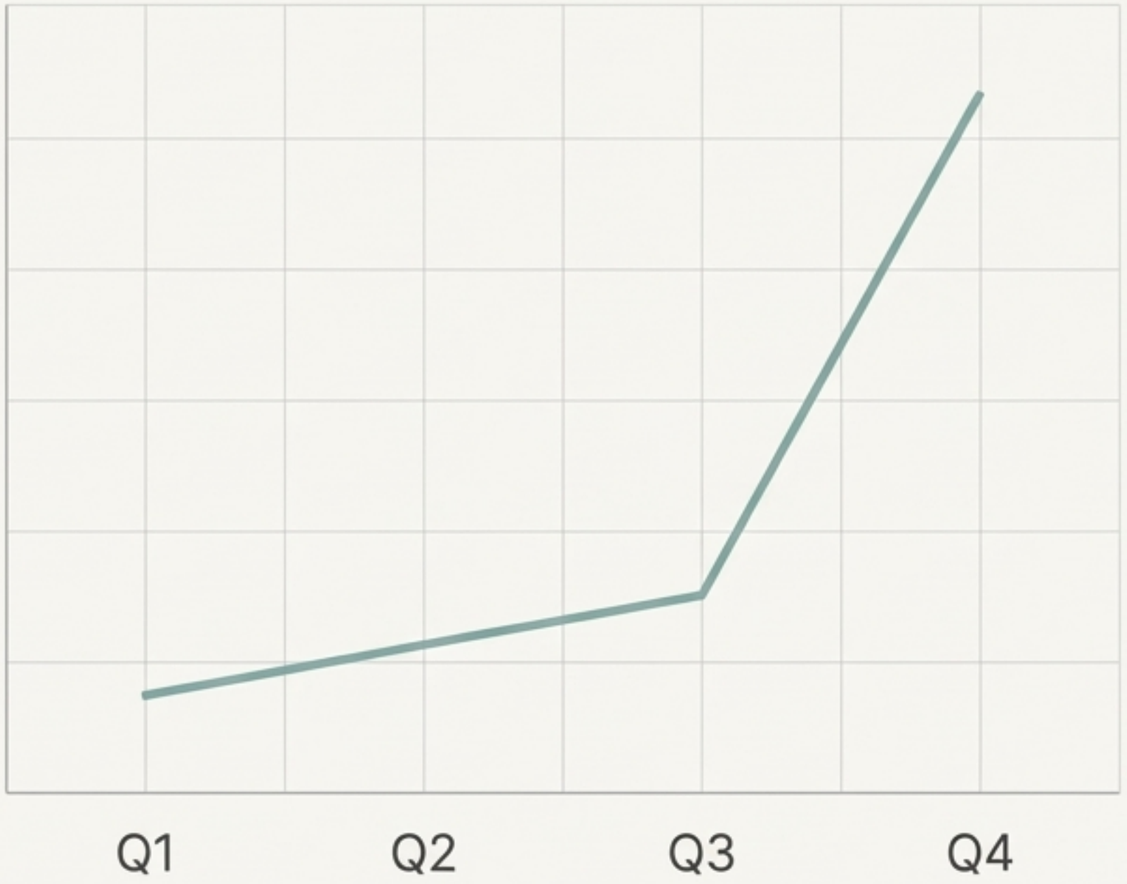


- **Insight:** High lift detected in matching theme or colour product ranges.
- **Takeaway:** Reveals highly profitable opportunities for targeted product bundling.

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Key Sales Analytics Insights

Indicative Results



- **Trend:** Sharp revenue peak in **October–November** (pre-Christmas).
- **Geography:** UK market dominates with **>80%** revenue share.
- **Operations:** Thursdays busiest; zero Saturday transactions.
- **Returns:** Cancellations impact **~2%** of total invoices.
- **Distribution:** Pareto pattern—a few products drive most revenue.

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Strategic Business Recommendations

1. Increase inventory forecasting aggressively ahead of the Q4 peak.
2. Create promotional bundles for high-lift matching product pairs.
3. Deploy targeted, automated win-back campaigns for At-Risk users.
4. Implement exclusive reward structures to protect Champion revenue.
5. Expand localized marketing in under-served European export regions.

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Challenges & Future Enhancements

Challenges

- Missing CustomerIDs constrained deep user analysis.
- Handling negative quantities required complex join logic.
- Strict query-first design limitations in Cassandra.

Future Roadmap

- Migrate to real-time ingestion (Kafka + Spark Streaming).
- Implement time-series demand forecasting (ARIMA).
- Deploy pipeline to cloud infrastructure (AWS EMR).

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From Raw Data to Actionable Strategy

Distributed ecosystems successfully transform raw retail transactions into precise business interventions.

Empowering mid-sized retailers with scalable analytics drives sustainable economic growth (SDG 8).

The floor is open for questions.

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